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Adoption of General Purpose Technologies and Inventions of Methods of Inventing

An Explorative Study of the Adoption of the Immersive Technologies Extended Reality and Digital Twins

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The Adoption of Technologies Defined as General Purpose Technologies and Inventions of Methods of Inventing

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Införande av generiskt användbara teknologier och uppfinningar som ändrar sättet att innovera

En explorativ studie om införande av XR och digitala tvillingar

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Sammanfattning

Denna studie ämnade undersöka lämpligheten hos de rådande teoretiska modellerna för införande av ny teknologi när det kommer till införande av revolutionerande teknologier som karakteriseras som generiskt användbara teknologier (GPT) och uppfinningar som ändrar sättet att innovera (IMI).

Med den ständigt ökande takten på teknikutveckling kommer vi sannolikt att uppleva mer frekventa implementeringar av GPT:er och IMI:er. Därför måste utövare ha rätt verktyg och strategier att ta hjälp av för att förstå hur man navigerar i sådana situationer och hur man bäst utnyttjar de möjligheter som en framgångsrik implementering av dessa tekniker innebär i organisationer och sektorers ekosystem.

Studien har visat att de rådande modellerna inte helt täcker och förklarar konsekvenserna som införandet av dessa typer av teknologi medför. De insikter som diskuteras i denna studie leder till skapandet av en kompletterande modell om teknologiinförande och kan ge en initial kunskapsbas för arbetet med att uppdatera eller utveckla nya modeller och teorier inom detta forskningsområde.

Arbetet i denna studie utfördes med ett explorativt tillvägagångssätt genom kvalitativa intervjuer med ämnesexperter där GPT- och IMI-koncepten undersöktes genom de två framväxande teknologierna, XR och digitala tvillingar. Studien presenterar också argument för att teknikerna, XR och digitala tvillingar, har potentialen att vara GPT:er och IMI:er. Dessa tekniker är dock fortfarande i en tidig fas med en framväxande karaktär där deras genomgripande inverkan ännu inte slagit igenom.

KTH Industrial Engineering and Management

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Abstract

This study set out to investigate the adequacy of the prevailing theoretical models regarding technology adoption when it comes to the adoption of revolutionary technologies characterised as general purpose technologies (GPTs) and inventions of methods of inventing (IMIs).

With the ever-increasing speed of technology development, we will likely experience more frequent implementations and adoption of GPTs and IMIs. It is not a one-off. Hence, practitioners must have proper tools and strategies to lean on for understanding how to navigate in such situations and how to best leverage the possibilities posed by a successful implementation of these technologies into organisations and industry ecosystems.

The study has shown that the prevailing models do not fully cover and explain the introduction consequences of these kinds of technologies. The insights discussed in this study lead to the creation of a complementary model about technology adoption and can provide an initial base of knowledge that can be further developed during efforts to update or develop new models and theories within this research area.

The work in this study was carried out in an explorative manner based on qualitative interviews with subject matter experts where the GPT and IMI concepts were investigated through the two emerging technologies, extended reality (XR) and digital twins. The study also presents the rationale that XR and digital twin technologies uphold the potential of being GPTs and IMIs. However, these technologies are still in an early phase with an emerging nature where their profound impact is yet to be seen.

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The thesis was done in conjunction with a larger study conducted by Ericsson Research which is presented in a series of reports called The Future of Enterprises. I want to express my sincere gratitude to the project team at Ericsson, Anders Erlandsson, Sarah Kenlind, Patrik Hedlund and Sepideh Matinfar and the rest of the Industry and ConsumerLab team for their willingness to collaborate and their inclusion of me in their team as well as their guidance and supervision during the research process. Their expertise and industry knowledge significantly influenced the outcomes of this study.

I would also like to extend my deepest appreciation to my thesis supervisor at KTH, Professor Mats Magnusson, for his invaluable guidance, expertise, and continuous support throughout this research journey. His insightful and constructive feedback greatly shaped the direction of this study.

To the informants who contributed to this study by sharing their expertise and insights through the interviews, I would like to express my gratitude for your trust and willingness to share your insights. Your participation has significantly impacted the quality and credibility of this research. While your identities remain confidential, please know that your contribution has not gone unnoticed and is deeply appreciated.

In conclusion, I am truly honoured and grateful for the opportunity to collaborate with Ericsson and for the guidance from Mats Magnusson who contributed to this Master's Thesis. Your support, guidance, and generosity have been instrumental in its successful completion. Thank you for believing in me and for making this research endeavour a truly rewarding experience.

Märta Ernfors

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1 INTRODUCTION

In this chapter, the practical and theoretical motivations for the relevance of this study are presented. The motivations are followed by the aim of the study together with the developed research questions to be answered.

1.1 Practical Motivation

Throughout time, innovation and technological advancements have played an essential role in the sustaining and development of the economic growth of societies (Schumpeter, 1927; Soete and Freeman, 1997). Bresnahan and Trajtenberg (1995) argue for technologies, such as the semiconductor, included in the concept of general purpose technologies (GPT) as a driving factor for periods of growth and technological progress. Similarly, Darby and Zucker (2003) argue that the concept of inventions of methods of inventing (IMI), such as the double-cross hybridisation for creating new varieties of hybrid corn, drives above-average productivity growth.

Could we now be facing a new era of technological progress and productivity growth thanks to the emergence of immersive technologies?

Immersive technologies are predicted to provide a huge growth opportunity for businesses and should therefore be of great interest to both researchers and practitioners across a wide range of sectors. For instance, the digital economy of today constitutes about 20-25% of the global economy and it is expected that the metaverse could in the future be an $8 trillion market opportunity (Sheridan, 2022). It is predicted that by 2026, 25% of the population will, daily, spend more than one hour in the metaverse (Gartner, 2022a) and by 2031, the revenue in the digital twin market is estimated to be $183 billion (Gartner, 2022b). Additionally, the XR market is estimated to experience a compound annual growth rate (CAGR) of 38.5% between 2022 and 2030 ending with a predicted market value of $394.8 billion (P&S Intelligence, 2022).

1.2 Theoretical Motivation

When introducing these new growth-increasing technologies, companies will potentially face challenges adopting them. As Kline and Rosenberg (2010) state, “the greater the changes introduced, the greater the uncertainty … of the organization to absorb and utilize the requisite changes effectively”. Technology adoption is a well-researched area (Venkatesh, et al., 2007) where many different models exist. Technologies that are those of a general purpose (GPT) and those that present a new method of inventing (IMI) are, however, complex in nature and deeper knowledge about them is still to be uncovered (Cockburn, et al., 2018). This study aims to contribute to the literature about the adoption of such technologies as well as suggest some practical implications for the handling of the adoption of these types of technologies.

1.3 Aim

This study aims to investigate the validity of the current frameworks about technology adoption when it comes to technologies classified as general purpose technologies (GPT) and inventions of methods of inventing (IMI). Does the adoption process of these types of technologies follow

the prevailing models and frameworks about technology adoption, or do these technologies place other demands on the frameworks? To understand this, three research questions have been investigated.

1.3.1 Research Questions

1. How is the adoption process of technology affected if the adopting technologies have characteristics of a general purpose technology (GPT) and an invention of a method of inventing (IMI)?

2. What are the characteristics of technologies with elements of a general purpose technology (GPT) and an invention of a method of inventing (IMI) that influence the technology adoption?

3. How does the industry context affect the adoption of these types of technologies?

2 EXPOSITION OF THEORY

In this chapter, the relevant concepts are presented as well as their interpretations and definitions that will be used throughout this study.

2.1 Technological Innovation

Technological innovations are by Branscomb (2001) defined as the successful implementation of technical ideas that are new to the specific organisation. There are two cooperating driving forces behind the work and introduction of successful innovations (Kline and Rosenberg, 2010). On the one hand side, innovations are demand-driven by market pull forces and respond to opportunities in the market. This sequence was for instance presented by Hollomon in which the innovation process is ordered as a linear flow, starting with a perceived need, followed by invention, innovation and diffusion or adaptation (Hollomon, 1965). On the other hand, innovations can also be driven by overall technological and scientific progress where opportunities for improving products' performances, reducing their costs or producing new products open up (Kline and Rosenberg, 2010). This is described, for instance, by the innovation sequence of Blackett (1968) which starts with developments in pure science, leading to advances in applied science, followed by invention, development, prototyping, production and sales (Blackett, 1968). Blackett's described process is therefore technology push driven, taking a starting point in technological advancements. A balance between these two approaches is required for successful innovation outcomes (Kline and Rosenberg, 2010). Hence, it is a complex matter of both "answers looking for questions and questions looking for answers" (Branscomb, 2001).

The two concepts mentioned in the introduction, general purpose technologies (GPT) and inventions of methods of inventing (IMI) are, in accordance with the sequence of Blackett, technological advancements with great potential of driving periods of technological progress and productivity growth. GPTs and IMI will later be described.

2.2 Technology Adoption

Technology adoption is well-researched, and many different models and frameworks exist that aim to provide an understanding of how technologies are adopted by the individual and organisations (Venkatesh, et al., 2007).

One important model is the diffusion of innovation (DOI) created by Rogers in 1962. The diffusion of innovation is the process of structural and/or functional social change, starting with a new idea and spreading across a unit of analysis. The model suggests that the diffusion of innovation is driven by the two-way convergence process of communication. It is influenced by various elements, being the innovation itself, communication channels, time and social systems (Rogers and Williams, 1983). The DOI is additionally focusing a large part on the different users and dividing them into innovators, early adopters, early majority, late majority and laggards. This grouping is based on when in time they adopt new technologies which is based on their different needs, perception of value and their different thresholds. The groups, as well as the influencing elements, are shown below in Figure 1.

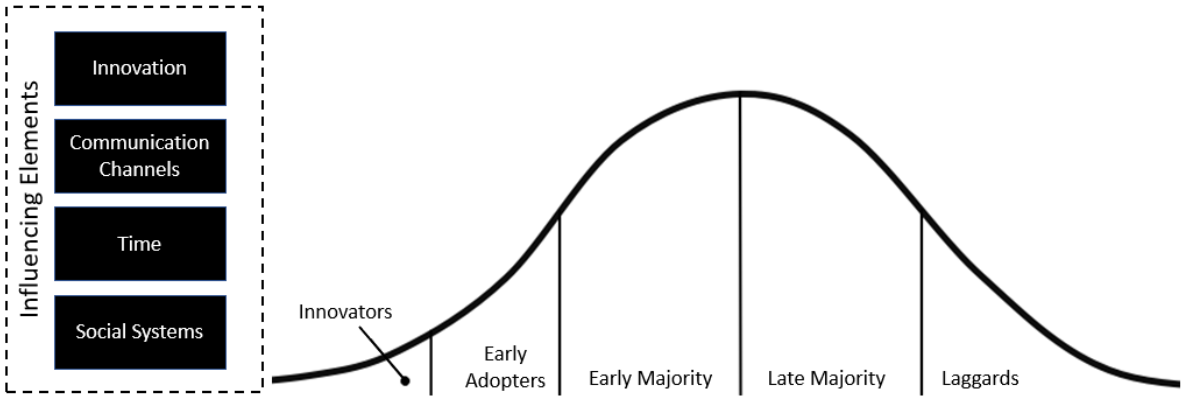


Figure 1. Illustration of the model developed by Rogers in 1962, the Diffusion of Innovation (DOI), as well as its influencing factors.

The technology acceptance model (TAM) developed by Davis (1986; 1989), is another adoption model but with a focus on the end-user's attitude toward the use of a technology. The user's attitude is based on the two parameters of perceived usefulness and perceived ease of use of the technology which are directly influenced by its design elements. The user's attitude toward the use of the technology is assumed to be a direct determining factor for the actual usage (Davis, 1986; Davis, 1989). The two parameters of the user's perception of the technology's usefulness and ease of use can be seen as an evolution to make the DOI model's two-way convergence process of communication more explicit. An illustration of the model is found in Figure 2 below.

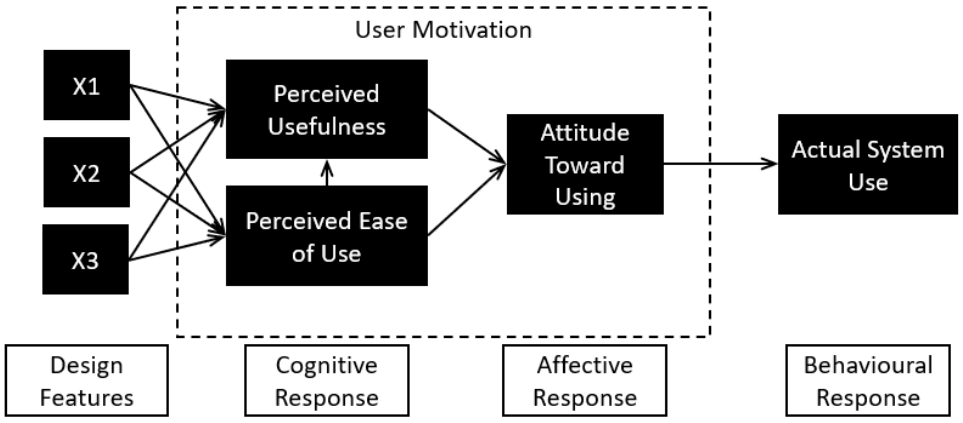


Figure 2. Illustration of the Technology Acceptance Model (TAM) developed by Davis (1986; 1989).

Another model, also with a focus on the user's attitude and behaviour, was constructed by Venkatesh, et al. (2003) as the result of an effort to unify several models within the research field of user acceptance. Eight models were reviewed and the result was the Unified Theory of Acceptance and Use of Technologies (UTAUT). The UTAUT model, as seen in Figure 3 below, is based on four factors highly impacting the user's behavioural intention and use behaviour; performance expectancy, effort expectancy, social influence and facilitating conditions. These impacting relationships are additionally influenced by gender, age, experience and voluntariness of use, referred to as key moderators (Venkatesh, et al., 2003).

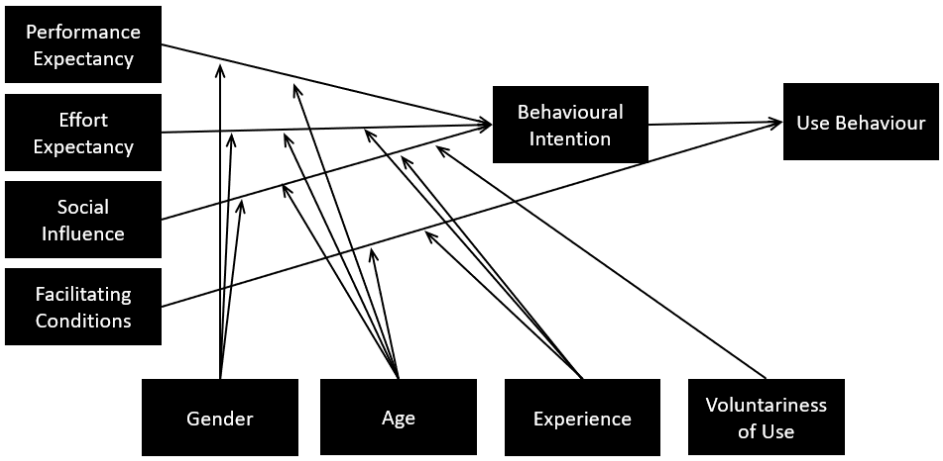


Figure 3. Illustration of the Unified Theory of Acceptance and Use of Technologies (UTAUT) model developed by Venkatesh et al (2003).

The model Technology-Organization-Environment (TOE), presented by Tornatzky and Fleischer (1990), investigates technology adoption with a focus on the company's context within which the technology is to be adopted. The framework divides the factors that influence technology adoption into three; the technological, the organisational, and the environmental context. These factors are interconnected and influence each other as illustrated in Figure 4 below.

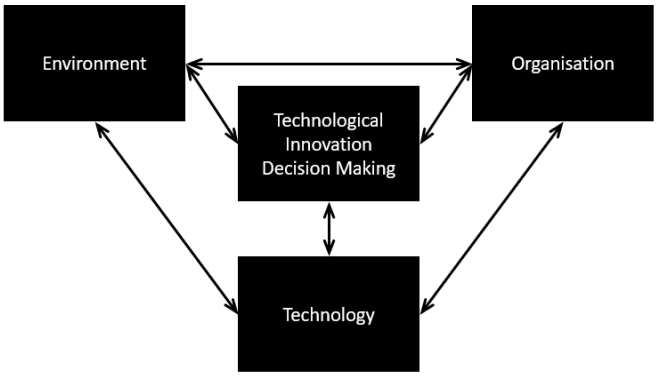


Figure 4. Illustration of the Technology-Organization-Environment (TOE) model developed by Tornatzky and Fleischer (1990).

In the technological context, influencing aspects are for instance characteristics of the technology itself but also other technologies already incorporated into the company or those in the company's external environment (Tornatzky and Fleischer, 1990). The organisational context includes several characteristics of the company adopting the new technology. This includes its size, the availability and quality of its human resources, as well as formal and informal structures and processes of the organisation (Tornatzky and Fleischer, 1990). The environmental context refers to elements of the external environment of the company such as its industry, its supply chain, customers and regulatory requirements (Tornatzky and Fleischer, 1990).

Most theories, herein the TAM and UTAUT, set out to explain the user behaviour or the individual motivation for the usage and acceptance of technology in organisations which

mainly helps understand the impact of technologies' design aspects (Li, 2020). While there are technological elements in both the TAM and the UTAUT models, covering; the performance level, usefulness, usability as well as effort needed to incorporate the new technology, these aspects are, however, discussed with an emphasis on the user's perception or expectancy of them. Of course, these are important aspects to keep in mind when introducing new technologies but in this study, the area of interest is how the adoption is affected by the characteristics of the technology itself. This is an area less examined by previous literature (Li, 2020), however, the DOI and TOE models partly take focus on the technology when understanding the adoption of technologies.

One question that emerges from these frameworks is whether these are completely comprehensive, covering all types of technology introductions. Technologies that are those of a general purpose and those that present a new method of inventing are complex in nature and deeper knowledge about them is still to be uncovered (Cockburn, et al., 2018). Does the adoption process of these types of technologies follow the prevailing models and frameworks about technology adoption or do these technologies place other demands on the frameworks? The current frameworks may have a tendency to overlook crucial impacts of technology characteristics on the adoption process.

To build further on the understanding of technology adoption in the future, there might be a need for updates to the current frameworks or the creation of new ones to additionally cover certain types of technologies. But in what way do they differ? Following this, the descriptions of the concepts referred to as GPT and IMI are presented.

2.3 General Purpose Technologies

Technologies that can be categorised as General Purpose Technologies (GPTs) are technological advancements that have a strong potential of being drivers of the innovation sequence as described by Blackett (1968). GPTs are considered "enabling technologies" due to their nature of fostering new opportunities and having the potential of pervasive productivity advancements throughout the economy (Bresnahan and Trajtenberg, 1995).

Based on a review of historical technologies that have had pervasive effects on the overall economy, as well as a summary of the state of the theoretical literature on GPTs, Lipsey et al. (1998) define GPTs with a number of technological characteristics. It is provided that almost all new technologies can retain one or more of the characteristics of a GPT but for a technology to be classified as a GPT all characteristics need to be obtained (Lipsey et al., 1998). Hence, according to Lipsey et al., which will be the definition used in this study, a GPT must:

1. have a wide variety of uses by being used in a wide variety of products, processes and activities, and/or constitute a central role in technology systems or main technologies.
2. be applicable across a wide range of uses where the use of the technology is spread across multiple parts/sectors of the economy.
3. have strong complementarities with other technologies.
4. have a scope for improvement and a process of evolution.

Characteristics 1 and 2 - “a wide variety of use” and “a wide range of use” - may, at first sight, seem similar. However, Lipsey, et al. (1998) give a good example where the difference is clearly shown:

> Electric lightbulbs and screw drivers are used more or less throughout the entire economy. Notice that having a wide range of use is not the same thing as having a wide variety of uses. For example, although it is widely used across the economy in many different settings, an electric lightbulb has only one use, to produce light. In contrast, lasers feed information to cash registers at store checkouts, provide precision cuts in surgical operations, read information from CDs, and print hard copies from computer outputs. (Lipsey, et al., 1998)

The technological complementarities, characteristic number 3, are the innovation-enhancing relations between the GPT and its downstream, user sectors. R&D efforts within a GPT are disseminated and magnified through its complementarities which creates new inventive opportunities in the user sectors. Simultaneously advancements in user sectors increase the incentives for technology upgrades in the GPT. This positive feedback loop propagates technology advancements throughout the economy (Bresnahan and Trajtenberg, 1995).

The scope for improvement, in characteristic 4, means that there is an evolution where e.g. the technology is improved, operation costs are reduced, the value increases thanks to the increase of supporting technologies/complementarities, the range of use widens or its variety of use increases.

Following this, Helpman and Trajtenberg (1994; 1996) have developed a framework for the specific adoption of GPTs - the diffusion of GPTs. This model investigates the different parameters that distinguish the early GPT adopters from the slower ones, the laggards or late adopters. Helpman and Trajtenberg present four parameters that are shown to play a key role in the adoption of a GPT and present an example of the parameters' impact on the adoption of the transistor within a few different sectors. The four parameters are:

1. The new GPT’s productivity advantage in relation to previous technology
2. The number of complementary components developed for old technology
3. Demand conditions or willingness to pay for relative advantage
4. The cost of R&D for new complementary components required.

2.4 Inventions of Methods of Inventing

Another classification of technologies with revolutionary impact is the concept of “the invention of a method of inventing" (IMI), coined by Griliches (Darby and Zucker, 2003) when describing the effects of hybrid corn. Hybrid corn was not an invention or improvement of a specific product that successfully could be adaptable instantly to all agricultural areas and their specific conditions. It was rather an invention that offered a new method of breeding new corn varieties and, in its extension, varieties of other crops. It was a new method of inventing (Griliches, 1957). Such “inventions of methods of inventing” are usually in the form of new methodologies or instruments and seldom in the form of new paradigm shifts (Darby and Zucker, 2003). According to Cockburn, et al. (2018), IMIs are inventions that offer new ways

to create new products or develop new research insights with broad applications and provide researchers with the foundation for developing their own approaches to innovation throughout various sectors. Hence, IMI encompasses novel methodologies, tools, and frameworks that enhance and accelerate the invention process itself. Cockburn, et al. (2018) provide an example of an IMI within AI:

> For example, while far from perfect, powerful algorithms to scan brain images (so-called functional magnetic resonance imaging [MRI]) have transformed our understanding of the human brain, not only through the knowledge they have generated, but also by establishing an entirely new paradigm and protocol for brain research. (Cockburn, et al., 2018)

As seen, the way in which IMIs appear, and will be considered in this study, can be both revolutionary as the basis for a new paradigm, such as functional magnetic resonance imaging, as well as more modest by presenting a new tool or methodology, however not without the potential of groundbreaking impact.

The width of impact in various innovation processes implies that IMIs potentially drive overall technological progress and affect the economy to a greater extent than a newly developed product as it opens for innovations through increased technological opportunity and/or appropriability for a wide range of different products and applications (Darby and Zucker, 2003; Cockburn, et al., 2018).

On the other hand, they potentially create major challenges for existing players. Companies organise their activities and build their knowledge and capabilities based on their experience of working with a specific technology (Henderson and Clark, 1990). Hence, new methods of inventing challenge the current ways that companies are organised and pose significant threats to established companies with well-established structures due to the technology's nature of revolutionising the methods of inventing (Rothaermel and Thursby, 2007).

3 METHOD

In this chapter, the methods used in the study are presented. The overall research design is presented, explaining the structure and design of the research process. This is followed by a presentation of the approach during data collection as well as the coding and analysis of data.

3.1 Pre-Study

The study's scope was the result of the insights gained from a pre-study where 11 subject matter experts were interviewed using semi-structured interviews. The informants in the pre-study are considered subject matter experts as they work within the area of new immersive technologies such as extended reality, virtual reality, augmented reality, metaverse, holographic technology, digital twins, etcetera. These informants were recruited using a global information services company with a worldwide expert network as well as through business contacts. The insights in this pre-study led to the delimitations and scoping of the study and understanding of the key topic. The insights from the pre-study also helped in narrowing down possible technologies that could operationalise the GPT and IMI concepts.

3.2 Research Design

The study seeks to examine the adoption as well as encountered barriers and enablers of new technologies covered by the two concepts, GPT and the IMI. Due to the topic's unexplored nature, the study took an explorative approach (Swedberg, 2020).

To obtain data on this topic without depending on informants being familiar with the concepts of GPT and IMI, two technologies were chosen to operationalise the concepts. These technologies were thoroughly studied to examine whether they indeed can be classified as GPTs and IMIs and therefore constitute a valid operationalisation of the concepts. Hence, two technologies were investigated, and the insights from them were then abstracted and synthesised into the GPT and IMI concepts.

The data were obtained through semi-structured interviews to understand the topic from experts and decision-makers within the field (Creswell and Poth, 2016). The semi-structured approach for the interviews enabled the discussion to dig deeper into interesting areas without restraining the conversation. It also enabled the informants to thoroughly discuss what they perceive as important aspects of the topic and their own experiences of it (Saunders, et al. 2015).

These qualitative research interviews (King, 2004) were the single source of data collection and the research method will therefore be of a mono-method nature where the qualitative data collection will be combined with qualitative data analysis (Saunders et al., 2015). Qualitative studies accommodate more flexible research and focus on novel discoveries (Bryman, 1984) which enabled interesting research aspects to be further developed during the course of the study.

An overview of the operationalisation of the concepts, GPT and IMI, to the technologies, extended reality (XR) and digital twin, is presented in Figure 5 below.

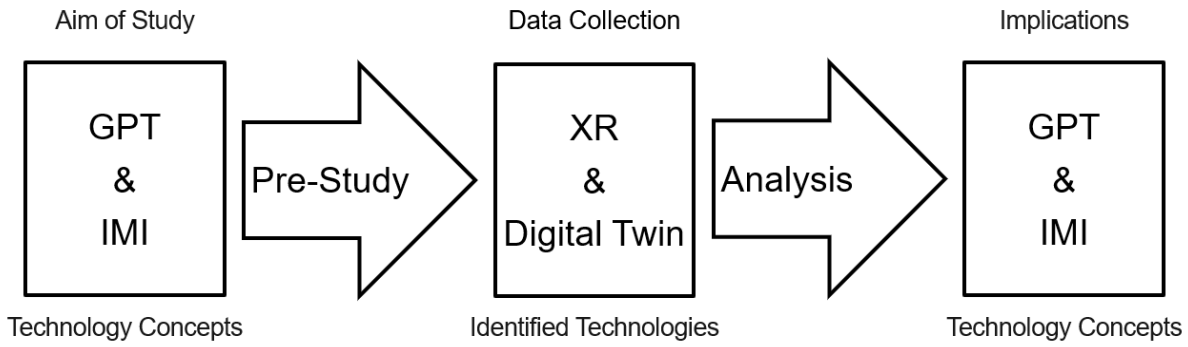


Figure 5. An overview of the operationalisation of the technology concepts, GPT and IMI, to the identified technologies, extended reality (XR) and digital twin.

Due to the newness of these technologies, and the fact that the study, to some extent, investigates expectations of the future, it is challenging to measure the technology itself without having to discuss expectations with experts. Due to this future outlooking nature of the study, the design was influenced by the Delphi study, where the opinions of experts are gathered to form a true expert consensus around the topic (Gordon, 1994). For this reason, subject matter experts were interviewed in this study in a semi-structured way to allow for deeper discussions with informants with valuable insights.

3.3 Data Collection

A total of nine interviews were conducted with a semi-structured approach. The informants of the study were subject matter experts and academic researchers, all with great insights into extended reality and digital twin technologies, their adoption in enterprises and their potential impacts. The informants are shortly presented as Informant A to Informant I in Appendix B. The informants were recruited through business contacts and the interviews were held in April and May 2023. The interviews were conducted digitally on Microsoft Teams with both audio and video. They were recorded with consent from the participant and the recording was later also transcribed. Additionally, the transcripts were grammatically modified for readability. The interviews were scheduled for one hour and followed a semi-structured interview guide which can be found in Appendix A. Due to the nature of semi-structured interviews, all discussed areas were not covered as thoroughly in all interviews but rather followed from the ongoing discussion in order to get qualitative, in-depth answers to the areas investigated.

The sample size was determined by saturation, hence, data was collected until no more new insights were found and data adequacy was reached which is considered an important part of qualitative research to obtain reliable results (Morse, 1995).

3.4 Data Analysis

The transcriptions were analysed, and summaries of important paragraphs and sections were created which were as true to the data as possible. The data was analysed, interview by interview, using the model of Gioia, et al. (2012). The summarised sections were deconstructed to identify the essence of the section and generate first-order concepts, per Gioia, et al.

The overall analysis was conducted with an abductive method due to the theory that was the base for this study, hence deductive, but with coding that was conducted as open and inductively as possible to identify interesting insights into the topic. The concepts were then given a second-order theme and further grouped into aggregate dimensions. Both the aggregated dimensions and the second-order themes of each transcript were revisited multiple times to ensure that all coding aspects were covered from all interviews due to the open approach to the coding. A full overview of the second-order themes and how they relate to the aggregated dimensions can be found in Appendix C.

Following the aggregated dimensions and the second-order themes, a coherent text was later composed using the first-order concepts to provide the analysed result comprehensively. This is found in *Chapter 5 Results and Analysis*.

4 Exploring Extended Reality and Digital Twins

In this chapter, extended reality and digital twin technologies are presented as they operationalise the GPT and IMI concepts in this study. Additionally, reasons for assuming that XR and digital twin technologies have the potential of being a GPT and an IMI are presented.

In order to study this topic, two technologies have been chosen to operationalise the concepts of GPT and IMI in the data collection. The analysis and insights will then be discussed in terms of GPTs and IMIs. The technologies chosen are the emerging technologies, extended reality (XR) and digital twins. A brief explanation of these technologies will be presented as well as a short argumentation for why these technologies can be classified as GPTs and IMIs and hence be used when discussing the adoption of GPTs and IMIs.

The ongoing technological shift to a so-called cyber-physical world with access to cyber-physical technologies and immersive experiences opens the possibility of merging the digital and physical worlds (Drath and Horch, 2014; Suh and Prophet, 2018).

Immersive technologies refer to the kind of technologies that creates a merging of the physical and virtual worlds, such as extended reality (XR), and allow users to feel engaged and immersed in their experiences and part of it (Handa, et al., 2012; Suh and Prophet, 2018). Studies have shown that utilising immersive technologies could, for instance, increase engagement in collaborative activities, improve efficiency and accuracy in task performance and increase creativity (Suh and Prophet, 2018).

Cyber-physical technologies, e.g., digital twins, support the integration of cyber and physical systems and elements. These system integrations could entail technological synergies, improved control and monitoring in real-time, and increased predictability and automation (Akanmu, et al., 2021). Extended reality and digital twins will be further presented in the following chapters.

4.1 Extended Reality

Extended reality (XR) is a hypernym encompassing immersive technologies in which the boundary between the actual reality and the virtual or digital settings merge in a combined environment (Alizadehsalehi, et al., 2020). It entails virtual reality, augmented reality and mixed reality technologies (Park and Kim, 2022; Alizadehsalehi, et al., 2020).

Virtual reality (VR) involves the use of a headset or a head-mounted display to immerse users in a fully simulated digital environment (Marr, 2019). By blocking out the real world (Alizadehsalehi, et al., 2020), VR can create a sense of presence and transport users to entirely virtual environments. It immerses the user in a 360-degree, responsive virtual world (Handa, et al., 2012; Park and Kim, 2022). Initially embraced by the gaming and entertainment sectors, VR has rapidly gained traction across diverse industries (Marr, 2019). Its potential applications have expanded beyond leisure, with numerous sectors such as engineering, construction, medical, the military, and others recognizing the immense utility of VR technology (Marr, 2019).

While VR presents a new reality, augmented reality (AR) enhances the user's experience based on the real-world environment where additional content, such as virtual objects and animations,

is presented as a layer superimposed on top of the real world (Suh and Prophet, 2018; Park and Kim, 2022). This can be achieved through devices such as smartphones or AR glasses which use cameras and sensors to blend virtual elements with the real world (Marr, 2019). AR reached mass market attention through the introduction of Pokémon GO where the animated monsters are overlaid on top of the user's real environment through smartphone cameras, as well as from the introduction of Snapchat filters (Marr, 2019).

Mixed reality (MR) is the experience of virtual elements in the real world that the user is also able to interact with in a realistic manner in real-time (Park and Kim, 2022; Alizadehsalehi, et al., 2020; Marr, 2019). It is the combination of VR and AR technologies in a seamless blend (Park and Kim, 2022; Alizadehsalehi, et al., 2020). MR experiences require the user to wear an MR headset (Marr, 2019).

The term "extended reality" is, therefore, the umbrella term that describes the continuum of experiences that range from purely virtual to fully augmented environments, encompassing VR, AR, and MR.

4.2 Digital Twins

A digital twin is the virtual or digital representation of a physical object, process or system (IBM, n.d.). The digital twin enables simulation of reality, for control and decision-making purposes, where its behaviour can be analysed and future scenarios predicted. These simulations are done in real-time with constant feedback and input of data from e.g. sensors (Tao, et al., 2019; Vatn, 2018; Akanmu, et al., 2021) to accurately reflect its physical counterpart. The insights gained from digital twins can then be transferred back to the physical counterpart (IBM, n.d.). The industries that mostly benefit from the use of digital twins are those with large-scale systems or projects with complex machinery or processes such as manufacturing, systems engineering, construction and production, design and power utilities (IBM, n.d.). The term "digital twin" was first used by Grieves in 2003 in the context of lifecycle management (Grieves, 2014).

One example of the utilisation of a digital twin is for Ericsson's radio base station sites. Instead of the previous manual-based approach to site inspection and site engineering, with digital twin technology, the sites with the radio towers and all components are, through the digital twin platform, visualised in a 3D environment with a 360-degree view of the whole site. The digital twin platform offers telecommunication service providers and tower companies new ways of capturing, connecting and utilising data from the sites with a data-driven and automated solution. The benefits of the digital twin technology are that it streamlines the network deployment process, ensures high quality and improves energy efficiency at the sites. Additionally, it enables better-informed decision-making for the design and upgrades of the sites, for the optimising of resource utilisation and for enabling the telecommunication service provider and tower companies to leverage the full potential of their networks. (Ericsson, n.d.)

4.3 Extended Reality and Digital Twins – Operationalised as GPT and IMI

Arguing for technologies' potential of being GPTs or IMIs is a complex task that requires thorough investigations of, for instance, investments and patents of the technology in various sectors. The reason for this is that it can provide evidence of the technology's potential for changing innovation processes and its general purpose (Cockburn, et al., 2018). Additionally, the categorisation of technology to be a GPT or an IMI is done ex-post (Darby and Zucker, 2003; Edquist and Henrekson, 2006) and it, therefore, need time in the market in order to show its impact in enabling technological breakthroughs. However, there are tendencies that imply that XR and digital twin technologies could be classified as such. In this chapter, the reasons for assuming XR and digital twin technologies have the potential of being a GPT and an IMI are presented.

4.3.1 Demonstrating a General Purpose Potential

Digital Twins

Digital twin technology is stated to be applicable in various industries where it is also said to be used to enhance operations in several different ways (Kan and Anumba, 2019). For instance, the initial use cases were found in aerospace. The National Aeronautics and Space Administration (NASA) and the US Air Force have applied digital twins to predict the remaining useful lifetime in complex systems of their vehicles (Tao, et al., 2019). Kan and Anumba (2019) also identified digital twins as critical for improving processes involved in a project's life cycle within the construction domain: the design stage, construction, operation and maintenance. Similar uses within design, analysis and control are also possible in the maritime and shipping industry (Semeraro, et al., 2021).

Tao, et al. (2019) additionally list different purposes that digital twin technology has, such as the design of production lines and production processes for manufacturing flexibility to improve performance and competitiveness. Digital twins also show great potential to be utilised in use cases for healthcare and injury prevention. In healthcare, a digital twin can for instance enable the safe testing of system changes within the hospital systems, predict therapy outcomes of patients and help determine health issues (Semeraro, et al., 2021). It can also prevent injuries from construction work by representing the worker as a digital replica where potential ergonomic vulnerabilities are identified. This is then communicated back to the worker with actionable insights through AR (Akanmu, et al., 2021). As a last example, digital twins have the potential to play a large role in city management with various tasks such as design of infrastructure and planning for efficient traffic flows, planning for crises and emergency evacuation, and also enabling sustainability and circular economies (Semeraro, et al., 2021)

Arguably, digital twins pose a great potential to have a wide variety of uses in a wide range of sectors where technological improvements in one sector possibly impact the development in other sectors as well as in the technology itself and its further development is yet to be seen. Hence, digital twin technology is in this study assumed to be a GPT potential.

Extended Reality

Similar to the digital twin, XR technologies display great potential for various uses in many different industries. For example, in their literature review, Suh and Prophet (2018) found evidence of XR usage for streamlined and increased performance in learning and education, pain, anxiety and stress reductions for patients through psycho- and physiotherapy, immersed user interaction for clothes fitting, and not the least, usage within gaming (Suh and Prophet, 2018). Furthermore, Gong, et al., (2021) present the range of potential uses of XR technologies within the manufacturing industry, being in the phases of design, assembly, operations, maintenance and training.

The same arguments used for the digital twin technology to have a GPT potential can correspondingly be used for XR technologies. XR technologies are widely used in several sectors where they are used in a variety of ways and with its widespread presence in multiple sectors, the technology will most probably exhibit strong complementarities with the technology-using sectors. Additionally, there is also great potential for further improvement and development of the technology.

4.3.2 Demonstrating a Method of Inventing Potential

Due to their general purpose with multiple possible uses, and the fact that these technologies are still early in their development, the reasoning regarding their IMI potential can be similar to that Darby and Zucker described for nanotechnology:

> At this early date it is speculative to identify the key invention or inventions enabling rapid burgeoning of nanotechnology. (Darby and Zucker, 2003)

But as Darby and Zucker continue, they believe it to be possible to determine a technology's "to be or not to be" before being shown after some time in the market. In the case of hybrid corn, the IMI and its effect were due to the creation of private incentives for new advancements, it was not a complete solution or product that would fit all geographical agriculture areas. Similarly, XR and digital twins have the possible effects of being visual aids, and inexpensive trials and errors enabling the creation of multiple new systems and products. It will be a new way to create and design smart cities or efficient product lines. It will not be a "one size fits all" development, but rather a way to create a system, structure or product that fits the specific requirements and conditions of the specific unit. Hence, in this study, XR and digital twin technologies will be assumed to exhibit the characteristic of an IMI.

5 RESULTS AND ANALYSIS

In this chapter, the results from the interviews with subject matter experts are presented. It is presented in accordance with the second-order themes and aggregate dimensions identified during the analysis. A full overview of the second-order themes and how they relate to the aggregated dimensions can be found in Appendix C.

Due to the inductive approach of the analysis, the coding was carried out without any specific framework or structure as a basis. The analysis resulted in four aggregated dimensions comprising a total of 22 second-order themes which were the outcome of the first-order concepts from the nine interviews. Each dimension from the analysis is given its own subchapter. These subchapters are the individual, the company, the industry and new innovation methods which presents the data gathered related to these dimensions. To the greatest extent possible, the texts in these subchapters are made up of their corresponding first-order concepts which are as accurate to the data as possible. After each concept, parentheses with letters indicate by which informant it was expressed. The informants are identified with a letter from A to I. A short presentation of each informant and their contributing role as informants in this study can be found in Appendix B.

5.1 Individual: Skills, Competence, End-User Focus

New job roles (B, C, D, F, G) and skill sets (B, D, F, G, I) will be needed in the future. The work will in the future be more of supervising the processes (G), understanding the patterns and insights (B), and experimenting with the data (B, C) and the new skill set needed might be more about AI competence (D) especially when efforts are made to make the systems smart to minimise the need for complex setups and reduce the competence need (G). On the other hand, a continuous need for upskilling is nothing new and not a big challenge (C). Technology, is technology, is technology (C). For technologists, it is still about being curious, experimenting, failing fast and learning new things (C). People need to feel that they have the right digital skills needed for their jobs (B) and it may require competence guidance to understand where the need for improvement is both for the employee and for the organisation (D, F). Stepping into digital twins is a pretty natural evolution and technologies built today should be sufficiently intuitive (H). People wouldn't need a lot of education, as long as they're not asked to do completely new jobs (H). People need not be trained more than gain an understanding of what is possible (F, H). What is happening a lot today is that people get new technologies but they're still working the way they did before. So, a little bit of training just to understand the tools and how to use them in the best way (H). It is not about training in the new technology. It is rather about training due to all of the routines of large corporations that will be changed (H).

It is a change journey for the organisation and people will need to evolve with the change (B, C). 10 years ago, people were afraid of robots in the production sites, now they are asking for more (B) and even though it will be more automated, it will still need to be human-centric (B, G) and the competence and technology need to be accordingly. But hopefully, people will become more productive and their jobs will get much easier. They should feel like super-humans (H).

The XR uptake is extremely slow. The number of VR sets sold in total is the same as the number of smartphones sold in 3 days (D). One of the reasons for this is stated to be that technology is not really mature. Technology needs to evolve and focus on the user (C, I). Devices are currently too bulky, and they should be possible to carry all day in the enterprise setting (C, D) and the glasses should be no more than 30-40 grams for that (D). They are also still too complex (E). When immersive technologies are getting into the hands of people and it's logical how to use them and it simplifies, that is when they will take off (E). High-tech technologies in the early stages don't take off. Some tech nerds might use it for a while but the big mass will not (E). History proves that if things get easier, smarter, or more efficient, we have a tendency to do it. If it increases complexity we don't do it. It takes off when it generates simplicity for you (E).

Because of the human-centric evolution of industrial work, an effort needs to be made on the technologies so that they focus on the end-user (C) with human-friendly interfaces where both connected and unconnected objects are present in the digital twin (G) to enable the human to troubleshoot (B), be aware of their environment (G) remote control in VR or AR (G), take decisions with vast amounts of data (G) and simplify for the users (E, G). Hardware needs to have a user-centric design that is slim, comfortable and possible to use all day without interfering with the performed task (C, D, E). Additionally, the XR experience, the content, needs to be made by storytellers who know how to convey stories, not by engineers as it is today (C). Because when you feel part of an experience, you get attached to it and learn faster (D, E).

5.2 Company: Process, Organisation, Change Management

For the organisation, the change management required for scaling is difficult because it means incorporating new ways of working (B, C). It will take them a long time to get smart about how to use these technologies because they will stick to old patterns of working and old processes and want to do things the old way (H) and it is not easy to move from hardware-business to software-business (I). Digital transformation is not digitalizing the analogue ways of working, but that's usually the first step conducted. It starts with digitalizing the current analogue ways of working because, of course, it's hard to design digital native processes when lacking experience with that (B). But organisations will slowly absorb these emerging technologies and internally change their workflows (D) and that means changing the current processes, ways of working and setup to do new things (B, C, I). Small companies that could change on a dime, could really benefit from these new technologies, take those extra steps and understand what questions to ask and what KPIs to create to really adapt the organisation to the technology (H, I). But for larger organisations, it is about the organisational complexity. It is not only about asking for new skills or competence development (B, C). It is up to the management to make the change in a controlled and orchestrated way and then to keep the people evolving with it (B). During scaling, more people with the right skills and competence will be needed and it will take time (B).

It is a transformational change because legacy systems need to be changed to be conducive towards digital twins and everything is tied together (C). These technologies bring enormous opportunities in multiple parts of the operations (B, H). However, the fact that the technology will be embedded in all organisational aspects - its wide variety - makes it complex to integrate

because all strategies need to align (B, C). For instance, the end-to-end production process needs to be visible and measurable in the digital twin and it should be clear where waste and outputs are (B). It should optimise the flow by enabling better and more intelligent decisions to be made (C). Sometimes really bad decisions are made because of different political perspectives or weak decision processes (F, H). It is surprising how little AR has progressed. But at the same time, the entire ecosystem and its legacies need to change to support this transformation and be conducive to the new technologies (B, C, H) and these legacy systems can be very hard to get out of (H). Organizationally, how does an organisation set itself up for these advanced technologies (C)?

People are up for working in VR, but it is up to the IT department to enable people to work in such environments (D). IT support is required in the whole change journey posed by the introduction of these immersive technologies (C). However, currently, IT capabilities are hindering the possibilities of experimenting and finding new uses for the data (B). Additionally, the sensitivity of data security and regulatory factors are limitations that make it harder to implement these technologies in businesses (E, F, I). In the industrial setting, there are concerns regarding the security and public exposure of mapped models of facilities or industrial plants (A). The safety and security of industrial companies are key (A). Mapped models of facilities and plants can never leave the firewalls of the company for processing in the cloud as this, for instance, can reveal intellectual properties and material flows (A). Adding to this, understanding the regulatory and legal effects or requirements of these new business implications is many times scarier than the new technologies as such (F). There is a lack of knowledge to be able to make a good judgement of, for instance, privacy aspects and other aspects that don't really have to do with what you are delivering (F). Despite this, it is important to experiment and figure out what more to use the data for (B, F) and that different players work closely together to explore the full potential of the technologies (F, G).

It is a change journey (B, C) and the change will never be a big bang (B). It is also argued that the impact on organisations will not be disruptive. It will grow slowly (E). And the only challenge to adoption is the tendency to over-hype a new technology shift, thinking it is much different when it just is a step-by-step improvement (E). These technologies are nothing new, they are just new names on the same, evolved, more powerful things with a more real-life feeling to them (E). When technologies have been talked about a lot, everything suddenly happens at a high pace - just like the ketchup effect. When the ketchup effect is happening, the technology gets easier to use (E). VR hasn't taken off yet because it is still too hard for the user to get started. But when it finally works, it has the best impact (E).

Furthermore, scaling the technologies is a lot harder than PoC:ing [creating a proof of concept of] an aspect of it because it is custom-made for every business (C). Industrial enterprises don't necessarily want to buy technology, they want to buy solutions, they have needs that need to be fulfilled (A). How do you build something that is custom-coding everything (C)? Digital twin vendors can not solve all problems. It is up to the organisation to manage many parts of the data, and that needs to be an intentional effort (C). On the other hand, when vendors are doing the first application, the cost is high, but when that is replicated for 1000 other manufacturing plants, then the cost is maybe just a fraction of the first implementation, and the company providing that application is making a huge profit (A).

Moreover, organisations need to be able to quantify the value brought by these technologies. Organisations don't approve scaling if there are no clear benefits shown such as time-saving

for people, product quality improvements or faster time-to-market (B, I). Implementing these technologies could, for instance, enable people to do more value-added jobs (B), enable flexibility while maintaining quality (C), save time, money and energy (F) or help you experience something which helps you make better and faster decisions (E, F). This is a type of agility that is worth money (E). It's a matter of finding those very good use cases. And they don't need to be flashy, they don't need to be mind-blowing, but they need to be really useful and hopefully also have a no-brainer return of investment (H). If it creates sustained value, who wouldn't want that (H)? But the value needs to be 100% tangible which they often are not, and then they are not measurable (D, I). If the value is not quantifiable as a top-line opportunity, the bottom-line cost will come out first (D). So it is an uphill battle comparing the measurable cost with the intangible promised benefit (D). Therefore, the number one challenge is always cost because it is easy to quantify (D, F, I).

Furthermore, introducing these immersive technologies can also enable organisations to quantify and increase the value of their offerings. Immersive technologies increase the value of an organisation's solutions because you can get a hands-on experience and connect emotions to it, hence a higher willingness to pay (E). If you can't measure, you can't act but if you can experience it in a digital representation then you can make real action (E).

5.3 Industry: Ecosystem, System Functionality, Industry Cycles

From the industry's point of view, the concept of a system functionality approach and dependencies were discussed. As an isolated factor, technology exists (C), but it is the combination of technologies which is going to change the world (C). It's a tool and a toolbox for people to build things (C). For example, XR is not big enough on its own, but in conjunction with other enablers - that is transformation (D). Hence, different actors need to work closely together to explore the technology's full potential (F, G, I). There are multiple components needed for these technologies to fly. For instance, there is a combination of software, hardware and killer apps that need to come together (A, E, I). Similarly, it is a chicken-and-egg situation (A, D) with three interrelated components: devices, networks, and content (D, I). Equivalently, the future of the industry is dependent on 3 As: AI, API and AR (D). How digital twins and immersive technologies relate and interoperate with AI should be top of mind for everybody right now (H). In the end, it is about how fast certain technologies are coming together, step-by-step, which will lead to a leap (E).

The fact that these technologies constitute a toolbox with a wide variety of uses, it is an unfair labour market where companies in various sectors are competing for the same competencies which are largely attracted by gaming companies (I). But on the other hand, companies can also learn from what the gaming industry has done and adjust the learnings to their own area, reused for other non-gaming applications without having to reinvent the wheel (I).

There is a great need for content (D). Applications are a hurdle (E) but all of a sudden comes a killer application, like Pokémon. But you never expect the change to come from there (E). However, currently, the available content is not good enough and there is a need for platforms that allow the creation of new volumetric content (D). There is also friction to download applications, but that's about to change with the Web AR/Web XR that allows you to open an AR session in a browser by just scanning a QR code, without the need to download an

application (A). Additionally, storytellers need to put the XR experience together to obtain an immersive experience. It can't be engineers as is today (C).

On the device side, the XR device market is dependent on the roadmap of one single vendor and is therefore taking baby steps (D). Currently, devices are too bulky (C, D). We have to put ourselves in the shoes of an end user (C), and the devices should be possible to carry all day in the enterprise setting (C, D) and with a battery life that allows that (I). The future of the devices needs to be comfortable glasses that don't get in people's way (C) and should be no more than 30-40 grams (D), or imagine when you can put in the contact lenses and start to get that type of immersive experience. That is just a technology barrier that doesn't work yet (E). Furthermore, in the industrial setting, there are issues with the device ecosystem (C). It is an existing absence of the industrialness or robustness of the devices (C) and the lack of existing compatible devices makes the planning of the technology implementation much harder (C).

Additionally, there is a need for low-latency networks to stream simultaneous data sources (F, G, I). The worker-centric Industry 5.0 increases the need for real-time components in the digital twin (G). 5G technology has room for improvement and a lot of potential that Wi-Fi and ethernet lack (C, D). Hence, industrial companies should invest more in 5G networks to accelerate the growth (I).

The magic with these technologies happens if you assemble enough use cases to justify a platform-thinking due to the fragmented market of the enterprise (D). There is a need for a platform to enable scaling within enterprises because yet it has not paid off to implement in scale (B). So there needs to be a priority on the data platform and connectivity (B).

Implementing a digital twin platform can create an ecosystem which will foster new innovation processes, just like the App Store, which disrupted the smartphone market (G). A digital twin platform for the on-top deployment of applications will simplify exposure to data and enable new entrants into the market (G). For instance, there exist intertwining efforts of digital twins to combine data and different data layers (G). An example of this is in a smart city setting where scattered data is generated from various non-compatible sources such as from different car vendors or mobile network operators. The intertwining of digital twins can then bring together different data sources and combine these different layers of data into one platform (G).

Similarly, there is a need for platforms that allow the creation of new volumetric content (D). These technologies are a tool and a toolbox for people to build things (C). When making things simpler and cost-efficient, these technologies will enable new types of innovations on top, as a new base, a platform (E).

Technology is not perfect yet. But slowly the experience is getting better (E). Headsets are slowly becoming smaller and we will surely start to see this being more integrated into our daily life (E). It is when advanced engineering is coming into the hand of consumers that things start to happen even more (E). For XR, consumers will be the big thing (D). When immersive technologies are getting into the hands of people and it's logical how to use them and it simplifies, that is when they will take off (E). Consumers have a big impact on industries. Industries now expect the experiences on the consumer side like the simplicity of an iPhone (E). On the other hand, consumer stuff might require more work to get it right and for that to really take off (H). However, sometimes the leap does not come from where you expect it to (E). Then all of a sudden, a killer application appears, like Pokémon, but it was never expected that the change would come from there (E).

Similarly, everyone has talked about AI for a long time but it was first when chatGPT came out and it became a hype that everyone had to jump on (D, E), but we already had similar solutions for a long time (E). When it has been talked about a lot, all of a sudden everything happens at a high pace - the ketchup effect (E). Quite often people overestimate how much the technology will change or improve in the short term and then underestimate it in the long term (I). The question is when this will disrupt on some other angle that previously has not been thought about (E). Because of all the potential opportunities in the markets, it would make sense to leverage some kind of industry consultant for understanding ways to improve and innovate to find new solutions to the current problems (I).

No sectors are currently seen as laggards in implementing digital twins (G) and because even conservative sectors such as manufacturing are showing interest, it gives good hope that there will be no more than a few sectors that don't have digital twins implemented within a few years (G). The widespread adoption will come by exploring faster sectors or low resistance paths, which drives additional needs through a snowball effect (G). You have to have the first product, the first killer application and find which segments that drive that adoption, and then it starts to expand from there and it draws other applications with it (A, I). Just make sure to have one happy customer, and then maybe 10 (H). And then figure out how things might scale (H). The fact that the technology can bring use cases in several sectors is an insane opportunity (H). Similarly, XR needs to have a general purpose otherwise the whole sector will be a complete commodity (D).

There is a lack of knowledge of the existing solutions (F, G, I) and the benefits of them (F). There need to be good portfolios for showcasing possible solutions (G, H, I) and proof points of prior deployments with tangible values (D). Otherwise, the whole sector is doomed (D).

People are up for using these technologies, but it is up to the IT department to enable it (D) but IT is dependent on procurement cycles of the industry (D). The different industry cycles are not aligned or synchronised, and the misalignment in technology readiness levels is an enormous challenge (D). However, now is a good time to disrupt the way we work and piggyback on the transformation that will happen, brought by the emergence of AI's large language models (chatGPT) which is now happening but has been lingering for decades (D).

5.4 New Innovation Processes: Culture, New Questions, Experimenting

These technologies can pose new innovation processes for the industry depending on how they are implemented (G). Digital twins could be a new way of inventing as it can influence the product development organisations' decision-making to also consider production factors in the design and development stages (C) and make mistakes early on - in the design stage rather than in implementation (F). It could be a new approach to innovation as it provides the data and parameters that enable people to have fun by experimenting and trying new things (B, F). It enables more of a playground or a laboratory (F) and that is what will drive innovation (B). Hence, there is a need to experiment and figure out what more to use all the available data to, and that will need to be enabled from the management's side (B, F, I). It starts from the company culture, where people are encouraged to try new things and fail in a safe environment (I).

Digital twins should also be simplified and a dimension of gamification should be added to it to facilitate scaling (G). Digital twins and all the technologies around them will help increase awareness and knowledge, and then hopefully also enable better questions to ask and make connections and dots where we wouldn't have made them before (H).

Probably the same way with XR technologies, that could open up a whole other field of inventions and innovations (H). Visualising something creates the next level of innovation (E). When interacting with the world around you in so many more ways, better questions can be asked that are not possible to ask today (H) and enabling an immersive experience can trigger new types of innovations since others can start to see and want to try new things. The more immersive technology you can get into the processes, the faster you can prototype and develop (E). If it simplifies, it can also generate new types of innovations - new paths to explore that you have never used before (E). When you make things simpler and cost-efficient, these technologies enable new types of innovations on top, as a new base, as a platform (E).

XR technologies as new innovation methods - that is really what the industry should be driving, and it is up to smaller companies and governments to lead the way and set new standards for new innovation methodologies and bring that into the enterprise and industry settings (D).

However, a new innovation approach is discordant with the current operations and a focus on finding new innovation paths needs to be in a pragmatic balance with the current operations and scaling (B).

6 DISCUSSION

As it crystalised in the analysis, the adoption of technologies being GPT and IMI are anticipated to impact on several levels: individual, organisational, and industry levels. These different levels will be discussed in this chapter along with a comparison with existing models. This is then followed by the implications of this study both for theory and practice, as well as proposed future research within this field of study.

6.2 How GPTs and IMIs Impact the Individual

When adopting a new technology, just as described in, for instance, the TAM and the UTAUT, the user's attitude toward the new technology is essential. The data give evidence that there needs to be a focus on the end-user and a human-centric approach to the development of these technologies and the implementation of them.

Upskilling the workforce with new technologies and formulating new job roles are still important areas. However, given the data, from a GPT and IMI point of view, these technologies do not seem to bring any specific critical need for workforce upskilling and employees' continuous learning. No specific new competence is seen to be needed from introducing these new technologies. There is rather a sustained need for general and continuous competence improvement in overall technology where people will be gradually evolving with the change.

Additionally, implementing GPTs and IMIs will require an understanding of how the technologies might change the current ways of working, how to utilise them best and what new questions to ask. The technologies in themselves are assumed to be sufficiently intuitive to understand and work with.

Building further on this, since these technologies provide a toolkit for new ways of working but also for new ways of inventing and experimenting in organisations, the technology must arouse the curiosity of the users.

The technology must enable and inspire the user to experiment with data and create different possible scenarios in order to leverage the full potential of the new technology. This will ensure a competitive edge, further streamline designs and processes, and foster innovations.

6.2 How GPTs and IMIs Impact the Company

In this chapter, two areas identified in the data relating to challenges companies encounter when faced with GPTs and IMIs are discussed.

6.2.1 Adapt the Organisation to the Technology – Not the Other Way Around

Technology characterised by being both GPT and IMI is likely to affect companies on a deeper level than the adoption of simpler technology. One of the most important implications is that it changes entire organisational processes fundamentally. As one of the informants so well described it from their perspective,

> Digital transformation is not digitalising the analogue ways of working, but that's usually the first step we do. We start digitalizing our current analogue ways of working because, of course, it's hard to design digital native processes if you don't have experience with that. (Informant B)

Initially, companies will, hence, adapt the new technologies to their current processes and ways of working when instead, they should be adapting the organisation to the technology.

Cohen and Levinthal (1990) discuss prior related knowledge as being crucial for understanding the value of new knowledge and efficiently leveraging it. Similarly is seen in this quote with the lack of prior experience. Though it is difficult to make use of new technologies if there is a lack of experience and prior related knowledge. For the best result, companies must be responsive to what best practices are presented by the new GPT and/or IMI and accordingly adopt new processes and habits based on the new technologies. New GPTs and IMIs can not be seen as a new additional piece in the organisational puzzle. They rather imply new puzzles for the organisation to be built, from the ground up.

The change brought about by the introduction of GPTs and IMIs is consistent with that of a change of transformational nature which involves a significant shift of a system and its structures and implies new understandings of processes and their purposes (Chapman, 2002).

Successful performance, just as emphasised by the informants in this study, requires an intentional effort, engagement and visionary leadership from management as well as to early on set the stage for, and anchor a new culture in the organisation (Head, 1997; Chapman, 2002). Additionally, just as the data show the importance of guiding employees to understand how to utilise the new technologies best, Chapman (2002) argues for reframing and clarifying the new role, relations and contributions of the employee and their work processes.

Then, to be able to sustain the performance of the implemented change, the employee aspect is discussed by Head (1997) where there, for instance, will be a need for sufficient training. This corresponds to the insights from the study emphasizing that it will be a change journey. It will not happen all at once and the employees must be allowed to continuously evolve with the change.

Due to the transformational nature of implementing these technologies, for a successful implementation, Chapman (2002) and Kotter (1995) highlight the influential change strategy of empowering people to act on the vision and encouraging risk-taking and self-creation and formulation of new solutions. When discussing IMIs, this approach is even more important. To be able to find new methods of inventing, employees must, in line with what is seen in the data, be encouraged to take risks, experiment and have fun with the possibilities that open up with the use of new technologies. Enable employees to formulate and continuously update their ways of working and innovation methods.

6.2.2 The Challenge of Calculating and Quantifying the Value of GPTs and IMIs

For companies to be able to implement these transformational changes, they need to be able to see the benefits, and the value needs to be quantifiable - as seen in the data of this study. As always, value is important but the characteristics of the GPT and the IMI place greater challenges on this. How do you calculate the value of a technology when it has a wide variety

of uses and the value is distributed across several different parts of the organisation? In addition, how do you calculate the value of a technology that creates new innovation methods? You have not received an innovation, you have only created the opportunity to create innovations. How do organisations put a number on this?

This corresponds to two of the parameters in Helpman and Trajtenberg's (1994; 1996) model of the diffusion of GPTs. Firstly, there needs to be a productivity advantage of the GPT compared to the prevailing technology and the second parameter to consider is the cost of required R&D for complementary components.

As proposed in the data, portfolios that present previous deployments with tangible outcomes and showcase potential solutions should be made available from the industry's side. This could enhance organisations' comprehension of the possibilities they can achieve with the new technology as well as assist them in calculating the potential business value and the productivity advantage in relation to the cost of deployment and complementary components.

6.3 How GPTs and IMIs Impact the Industry

In this chapter, three areas identified in the data relating to the impact of GPTs and IMIs on an industry level are discussed.

6.3.1 A Holistic Approach, with Network Effects and Systems-Thinking, is Needed to Understand the Development and Adoption of GPTs and IMIs

In accordance with the GPT characteristics being its wide range of use and its complementarities, the concept of network externalities will have to be taken into account.

Network externality means that the value and the benefits of the good increase with the number of users (Katz and Shapiro, 1985) - the installed base. The increased size of the installed base increases the demand for complementary goods. In turn, the existence of complementary goods can influence the demand for the primary good (Katz and Shapiro, 1985). This network effect has been noticed by the informants, where discussions are held with regards to exploring sectors seen as low resistance paths or low-hanging fruits which will drive further demand and interest from other sectors. The value of the technology is indirectly increased by the number of users as it increases the amount and variety of complementary technologies.

Adjacent to this discussion is the chicken-and-egg metaphor as raised in the data. The growth and development of one technology increases the need for the development of other key components. The success of these technologies is dependent on systems thinking (Arnold and Wade, 2015) where multiple technologies and actors work closely together, combining their functionalities and efforts to explore the full potential of the technologies. It is the combination, the conjunction of technologies that is going to change the world. That is when a transformation can happen. Hence, new technologies characterised as GPTs will require systems thinking and consideration of network effects, because implementing such changes is more effective with a holistic approach where the whole ecosystem is connected and collaborate.

6.3.2 Drive Innovation in a Fragmented Industry Market with a Platform Approach

Due to the wide range of uses and the utilisation of a digital GPT across industries, a platform approach offers numerous benefits for the industry. These include simplifying data exposure, supporting content and application creation, and driving continuous innovation. With connectivity as its backbone, a digital platform approach fosters network effects that encourage collaboration among various players in the ecosystem. This expands the network of partners involved, embracing diverse players with different goals and capabilities (Luo, 2022).

A platform simplifies data exposure for companies, collected from various sources, and enables new entrants to access it easily. By providing standardised infrastructure, tools, and APIs, the development process becomes more straightforward for others. This not only facilitates market entry but also empowers developers, startups, and established companies to build on top of the platform and create new applications or services.

Moreover, in a fragmented market, a platform approach plays a vital role in enabling the rise of small and agile global players. These players leverage the platform's capabilities and opportunities to develop unique value propositions (Luo, 2022) and can successfully leverage the new methodologies toward innovation posed by the emergence of an IMI.

Through the platform's offering of existing infrastructure, tools, and services, businesses can reduce development time and costs. This efficiency allows for the allocation of resources towards creating new value propositions on top of the platform, further fueling innovation.

Embracing a platform mindset establishes a strong foundation for continuous growth and helps businesses maintain competitiveness in the fast technological development of society. By fostering collaboration, enabling experimentation, and integrating various technologies, a platform approach becomes a catalyst for driving continuous innovation and staying ahead in the rapidly evolving market.

Thanks to the possibilities enabled by a digital platform as well as the wide range of uses and the spread across sectors of the GPT, it is not evident where the technology will see its breakthrough. Due to this, the leap may not come from where it is expected.

6.3.3 Misalignment of Industry Technology Cycles – A Barrier to Technology Uptake

For complex technology introductions, as new GPTs and IMIs are, it is critical to keep the industry and technology cycles in mind and use that as a basis for introduction and adoption strategies. Understanding and analysing industry cycles provides a valuable perspective to explore the correlation between the nature of industry cycles and sectoral innovation patterns (Tan and Mathews, 2010). For instance, innovation efforts should occur during periods of downturn where costs are low and technology accessibility is high (Tan and Mathews, 2010)

On the other hand, technology cycles follow from the emergence of a technological discontinuity where a technology is substituted with a new one (Anderson and Tushman, 1991). Before a dominant design can emerge with a clear course of incremental technological developments, the new technology is challenged by major developments in the prevailing

technology and then followed by a process of refining the first crude introductory technology (Anderson and Tushman, 1991).

The industry cycles are not necessarily synchronised with the technology cycles. Hence, if the maturity of the technology cycle of the GPT and/or IMI misaligns with the industry cycles of its users, i.e. the downturns when there is time to invest in new R&D efforts and upgrades, the uptake of profound and permeating technologies might be low.

6.4 Comparison with Existing Models

When it comes to discussing the insights found in this study together with the existing theories on technology adoption, there are clear similarities with previously presented models. As a start, the TAM and the UTAUT models have a strong focus on the individual's perspective, emphasising the importance of aligning new technology introductions with the users and getting them on board. A similar discussion is held in this study both by having a human-centric approach as well as the anchoring of new culture and clarifying the new role of the employee for a successful implementation of a transformational change that is the result of GPT and IMI adoption in organisations.

Nevertheless, the individual perspective is not sufficient. As seen in the TOE, the adoption of new technology into the organisation is dependent on the technological, organisational, and environmental aspects. The organisational and environmental aspects of the TOE can roughly be equated with the company and the industry levels discussed in this study. Regardless, there should be an ask for a more thorough emphasis on transformational change management in the organisational factor as discussed in this study on the company level. There needs to be a major focus on the altering of new processes in organisations.

However, judging from the insights and discussion of this study, the TOE's technological factor is not sufficient to stand on its own when it comes to the adoption of GPTs and IMIs. When adopting such profound and permeating technologies, the technological aspect needs to be considered on all levels of impact. Just as the discussion about the company having to adapt to the technology, so too should the theory. The technological aspect should be infused in all adoption aspects discussed. The technology permeates all levels and can not be analysed on its own, as done in the TOE model. Technology can not be seen as a separate factor in the model but rather underlying the other factors influencing the adoption. In addition, the different levels influencing and affected by technology adoption are also interrelated with interdependencies. Based on what has been evident in this study, Figure 6 below shows a new proposed model complementary to the current theories about technology adoption, which helps to understand the GPT and IMI adoption process.

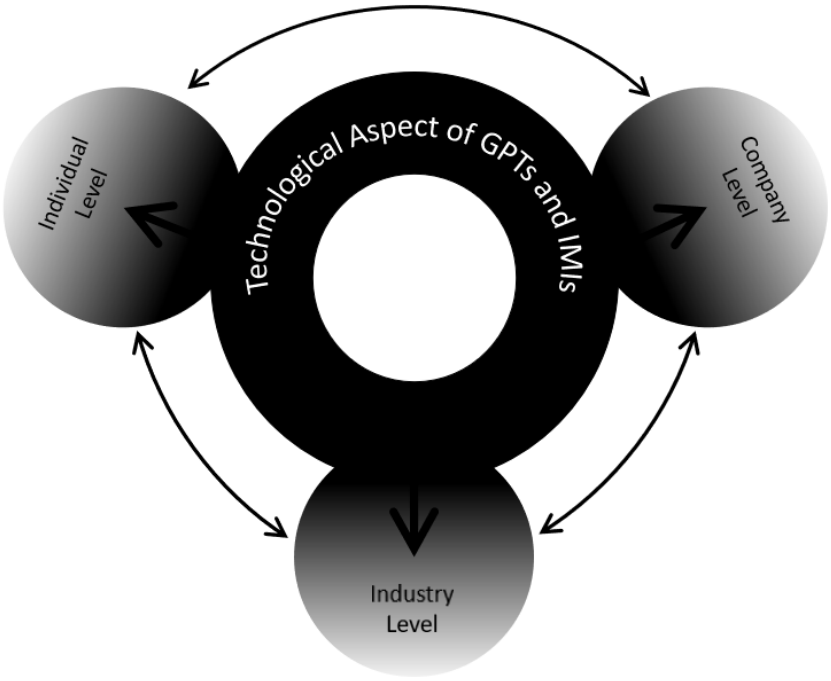


Figure 6. Illustration of the proposed complementary model for GPT and IMI adoption where the technology is infused in the different levels impacting and affected by the technology adoption.

Hence, it is not the technology in itself but rather its impact on the individual, company, and industry levels that need to be considered when modelling the adoption process of a GPT and IMI.

To understand the process, there is a need for a holistic approach involving the width and depth of the technology's impact on the individual, the company and the industry level. To manage this impact, organisational and process flexibility needs to be incorporated into the structures of companies and industries.

It will also be vital to engage with various stakeholders, which includes industry leaders, experts, employees, and other actors in the industry, to understand their perspectives and how to collaborate successfully. This also helps identify potential barriers, risks, and opportunities associated with the new technology implementation.

Moreover, there will be a need to investigate and develop an understanding of relevant and necessary policies and regulations to ensure safety and security for all players affected by and involved in the new ecosystems created by the introduction of GPTs and IMIs.

All these aspects must be covered and this arguably requires a systemic approach. Therefore, the adoption models must leverage systems theory and systems thinking to a greater extent due to the complex interdependencies within the system and its components. This is a research area where insights are lacking, and future studies are needed to understand this phenomenon in-depth.

Furthermore, for these kinds of technology introduction into organisations, the industry actors affected by this need to understand the impact of the synchronisation of the technology cycle and the industry cycles and how they affect each other. This has not been thoroughly addressed in previous adoption models.

6.5 Theoretical Implications

This study set out to question the adequacy of the prevailing theoretical models regarding the adoption of technologies characterised as GPTs and IMIs. It has been shown that the prevailing models do not fully cover and explain the introduction consequences of these technologies at the individual, company and industry levels. The insights discussed in this study can provide a base for understanding what to investigate further during efforts to update or develop new models and theories.

This study has also contributed theoretically by presenting the occurrence of the complex interdependencies between different aspects, other technologies and system components that need to be taken into account in the adoption process of new technologies. The adoption of technologies characterised as GPTs and IMIs is a challenging process where systems theory and systems thinking need to be leveraged to a greater extent as discussed in *Chapter 6.4 Comparing with Existing Models*.

One additional theoretical contribution of this study is the brief argumentation for the GPT and IMI potential of XR and digital twins. Yet, to completely determine whether these technologies with confidence can be categorised as GPTs and IMIs, a thorough literature review of the patents and the applications of the technologies needs to be conducted. This will sort out if there is a wide spread of application sectors and a wide variety of uses, as well as if it enables new ways of inventing. However, due to the emerging nature of these technologies and the lag in publications, these data points might not yet be visible and available.

6.6 Practical Implications

The insights gained from this study, impact businesses in several industries and on various levels. By understanding these technologies' width and depth of impact, companies and industry actors can make more informed decisions and create better-suited plans and strategies when introducing new profound and permeating technologies.

As technology continues to advance at an unprecedented pace, we will likely experience more frequent technology implementations of these sorts. It is not a one-off. Hence, practitioners must have useful tools and strategies to lean on for understanding how to navigate in such situations and how to leverage the possibilities posed by the successful implementation of these technologies into organisations and industry ecosystems.

There are several insights that managers and decision-makers can benefit from. For instance, management must pursue a broader approach to the adoption of technologies characterised as GPTs and IMIs, taking the ecosystem and platform thinking into account and following the guidelines of transformational change.

Lastly, on the individual level, users should continue being curious and explorative when using new technologies and aim to find new ways to better utilise them.

6.7 Limitations

Limitations in this study include the limited number of technologies that could be investigated (XR and digital twins). If a larger scope had been possible, more technologies classified as

GPTs and IMIs could be investigated. Furthermore, GPTs and IMIs were investigated together in this study. If they would have been studied on their own, further implications derived from only one of them could be found.

Moreover, the fact that XR and digital twins still are quite new technologies could impact the results where the informants (despite being subject matter experts) do not have so much experience in the actual introduction and adoption of these technologies in an enterprise setting.

Additionally, the chosen subject matter experts could have a biased view of this area as they are well acquainted with these technologies.

6.8 Validity and Reliability

For validating the use of XR and digital twin technologies to operationalise the GPT and IMI concepts, a chapter about why these technologies demonstrate a GPT and IMI potential is created. In this chapter, arguments are provided to prove that these technologies can indeed operationalise the GPT and IMI concepts.

The single source of data, consisting of the qualitative interviews, could imply reliability deficiencies. However, following the purpose of a Delphi study, the value of the study is not to provide statistically significant results but rather to generate ideas and informed insights with fewer informants or respondents (Gordon, 1994), which is a suitable approach due to the explorative nature of the study. Furthermore, to mitigate the potential reliability deficiencies in the data, data was collected from interview informants until saturation was obtained and no significant new insights were found.

Furthermore, the abductive approach in studies implies risks with personal biases when analysing the data. To reduce this risk, the analysis was conducted as openly and objectively as possible, where the transcripts were revisited multiple times to allow for overlooked insights to be found.

Additionally, questions regarding the generalisability of the results may emerge. These can be questions such as if there are companies or industries where the insights from this study are not valid and if so, is that dependent on, for instance, the number of complementary components developed for the previous or current technology (the prevailing technology before the GPT) as discussed by Helpman and Trajtenberg (1994; 1996). Could it further be influenced by the technological skills and competence of the individuals, the size and complexity of the company, the number of players in the industry, and the technology maturity of the company or the industry?

As this is an explorative study, it aims to give a first glimpse into this area and provide a base of initial understanding for future studies. Hence, how these aspects influence the generalisability of the results in this study is recommended to be investigated in depth in future studies.

6.9 Future Studies

Because of their complexities and their width of impact which are still not fully understood, technologies characterised as GPTs and IMIs are research areas that need to be further studied.

GPTs encompass a wide range of applications and capabilities, making them a complex research subject. Its ability to perform diverse tasks introduces challenges in understanding and predicting its potential impacts. Additionally, GPTs' distributed nature adds another layer of complexity to research efforts. Exploring the dynamics of GPTs' development, deployment, and adoption becomes crucial to gain comprehensive insights into its implications for the individual, the company and the industry.

Similarly, IMI presents unique challenges in research due to its profound influence on innovation processes. Investigating the impact of IMI on collaboration dynamics, how these new approaches integrate with existing practices and processes, and the broader innovation ecosystem is essential for researchers seeking to grasp the implications and potential of this field.

In this study, it has been shown that the current adoption models are insufficient for understanding the adoption of GPTs and IMIs. However, further research needs to be conducted to update or develop new models and theories covering the full aspects of their impact. Adding to this, the aspects that might influence the generalisability of the result in this study, as presented in *Chapter 6.8 Validity and Reliability*, are also recommended to investigate further.

Furthermore, in this study, GPTs and IMIs have been investigated together. For a better understanding of the concepts, they should in future research, be investigated individually, separated from each other.

With the ever-increasing speed of technology development, research insights are needed to understand the frequency of GPTs and IMIs. More research is needed to investigate the occurrence of GPTs and IMIs in history, in the present, and a prediction about the future.

Additionally, a discussion has been held on the platform approach for GPTs and IMIs. Future research can investigate how a platform and the increased amount of strong interdependencies in the ecosystem affect the adoption of GPTs and IMIs. Furthermore, more insights are needed with a systemic approach. This will facilitate the creation of best practices and strategies to leverage for successfully adopting these kinds of technologies in this setting.

7 CONCLUSIONS

In this chapter, conclusions from the discussion carried out in the study are presented along with closing remarks on the research questions. It is organised according to the three research questions in the respective subchapter.

With the ever-increasing speed of technology development, we will likely experience and need to be able to manage more frequent technology implementations. Practitioners must have useful tools and strategies to lean on in order to navigate and build an understanding of how to best leverage the possibilities posed by successful implementation of new revolutionary technologies. By this, companies and industry actors can make more informed decisions and create better-suited plans and strategies when introducing new profound and permeating technologies.

7.1 How the Adoption Process of Technology is Affected if the Adopting Technology Have GPT and IMI Characteristics

The change brought about by the introduction of general purpose technologies (GPTs) and inventions of methods of inventing (IMIs) is consistent with a change of transformational nature which involves a significant shift of a system and its structures. It further implies a new understanding of processes and their purposes. Therefore, companies need to be responsive to the best practices presented by these technologies and adapt their organisation accordingly instead of trying to fit the new technologies into the current structures and ways of working.

To implement, companies will need assistance in understanding the benefits of implementing new technologies of these kinds. GPTs and IMIs are complex concepts, and their effects and impact on companies are no less complex and challenging for companies and practitioners to understand. To facilitate companies in this matter, technology suppliers and other industry actors should leverage comprehensive portfolios and proof points of previous deployments for showcasing opportunities with tangible outcomes. This can then enhance organizations' understanding of the opportunities they can unlock with the latest technology, helping them evaluate the potential business value relative to the expenses of technology implementation.

Based on this, a complementary model to the existing theories about technology adoption was created. As seen in prevailing technology adoption models, the introduction of new technology in the company setting is impacted by various aspects. This study has identified similar factors as in previous models, which are considered crucial for the adoption process, herein referred to as the individual, the company and the industry levels combined with the technological aspect. All these aspects exist in previous models, although in different forms. Moreover, a key element is the emphasis that the technological aspect cannot be seen as an isolated aspect. It is underlying and permeates all other aspects that must be considered when investigating and building an understanding of the adoption process.

7.2 Which Characteristics of GPTs and IMIs Influence the Technology Adoption

GPTs and IMIs are complex concepts, and their effects and impact on companies are challenging to understand due to their profound and permeating nature. It is the technologies width and depth of impact on several levels which majorly affects and places greater demands on the process of technology adoption. An additional influencing factor on the adoption process is the transformational nature of the change that the technology implementation entails. Furthermore, their strong complementarities with other technologies and their new methodological implication are demanding and affect the adoption process where a better understanding needs to be obtained on systems theory, systems thinking and how the whole systems, processes and structures will be impacted.

7.3 How the Industry Context Affects the Adoption of GPTs and IMIs

On the industrial level, emphasis is put on ecosystem relations and the different industry actors that need to work together to leverage the full potential of the technology. The adoption of technologies characterised as GPTs and IMIs is a challenging process where systems theory and systems thinking need to be leveraged due to the strong complementarities with other technologies and their new methodological implications. Additional emphasis is placed on platform thinking that needs to be leveraged to get the most out of these technologies. Furthermore, there is a significant challenge at the industry level. The cycle of the industry, with its periods of up- and downturns, and the technology cycles are currently not necessarily aligned. However, an alignment is crucial for the successful adoption of technologies of these kinds.

To conclude, as GPTs and IMIs are likely to emerge more frequently, organisations and industry ecosystems must be more flexible with the ability to change. They must be able to adapt their structures and processes to the technology - not the other way around. It is a challenging approach that has to be accepted where major elements of systems theory and systems thinking need to be leveraged to a greater extent due to the complex interdependencies between different aspects, other technologies and system components. There must be a constant readiness to rethink the way things are done.

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APPENDIX A: INTERVIEW GUIDE 1(2)

Introductions (5 min)

- Thank the informant for being part of the interview
- Introduce myself and the study
- Material will be treated confidentially. Is it okay to record?

Company and Informant Profile (5 min)

- Can you please tell us a bit about yourself and your background

Views on XR and Digital Twins

If they describe how they see XR/digital twins and how they think they will be used, then you can tell if they see them as GPTs and/or as IMIs.

- How aware are you of the usage of XR/digital twin in businesses today?
    - To what extent are they being used?
    - What industries are leading the uptake of these technologies?
- In what type of business do you foresee the greatest benefits from the use of these technologies? Why?
- In what way, do you think, the use of XR/digital twin technologies will change/advance/evolve in the near future?
    - Is it (or will it become) a technology with more of a general purpose or is it specified to a few single tasks/activities?
        - What are those tasks/activities?
    - In what parts of the organisation will the technology be used?
    - Will it be used as an underlying tool for inventing and creating innovations?
        - Will it impact the innovation method/process?
        - In what way?

Views on Adoption

- What would be required from organisations to be able to leverage these technologies (to a greater extent) in their organisations?
- What are your general thoughts on the <u>challenges/opportunities</u> when introducing a technology like this?
    - If not mentioned, probe on 1,2,3.

1.
    a. What is your thinking regarding organisational needs for new competence and/or upskilling in order to make use of these new technologies?
        i. What challenges do you foresee here?
2.
    a. What impact do you think these technologies will have on companies' ways of working?
    b. In what ways do you think they will impact how organisations are organised?
    c. What impact will the new technologies have on the innovation method/process?
    d. Will the introduction of this technology change the way companies innovate/invent?

APPENDIX A: INTERVIEW GUIDE 2(2)

3.
    a. What would be the essential customer values and benefits when introducing these technologies?
    b. What are your thoughts on these technologies' potential to help offer new products and services to customers?
    c. Do you have any thoughts on new ways that value could be created because of these technologies?

Probe for GPT vs IMI:
GPT:
Do you see this because technology can be used in a wide variety of areas of use?
Do you see this because technology can be used in a wide variety of tasks?

IMI:
Do you see this because technology enables new ways to innovate/work?
Do you see this because technology revolutionises the innovation/working process?

Conclusion (2 min)

1. Are there any relevant questions that were not asked during this discussion?
2. Thank you & goodbye!

APPENDIX B: PRESENTATION OF INFORMANTS 1(2)

Informant A

Informant A is the former CEO of a company developing AR technology for mapping physical spaces and combining them with digital content. Informant A is based in Finland and has a background in startup companies, material science and drones. Informant A contributes to this study as a subject matter expert on XR and immersive technologies.

Informant B

Informant B is the head of a large factory in Estonia. The main assignment of the site is the development of new product introductions meaning they make them technologically mature to then be produced in partner factories. This means a low volume-high mix of immature products. Informant B has a good understanding of what Industry 4.0 means in practice. Informant B is in this study considered a decision-maker and a subject matter expert on the introduction of immersive technologies in an enterprise setting.

Informant C

Informant C is a US-based Industry 4.0 expert in a smart factory setting and is responsible for the development of Industry 4.0 use cases and physical automation. Informant C has been part of several proofs of concepts of new immersive technologies in the enterprise setting but is now more focused on the scaling component in this setting. Informant C is in this study considered a decision-maker and a subject matter expert on the introduction of these technologies in an enterprise setting.

Informant D

Informant D is a Visiting Professor at King's College London and is a Fellow of the IEEE, the Royal Academy of Engineering, the Royal Society of Arts, and the Institution of Engineering and Technology. Informant D is currently based in the US, engaged with the US and UK governments and sits on two advisory boards, the FCC, the US regulator, and Ofcom, the UK regulator. Informant D is for this study a subject matter expert on immersive technologies and engaged with the integration of 5G technology into emerging XR propositions.

Informant E

Informant E is the CEO of a company of sustainable innovation and green-tech investment subsidiary of a global leading company in smart digital reality solutions. Informant E has previously been involved in many industry introductions of new technology and industry connectivity. Informant E is based in Sweden. In this study, Informant E is considered a subject matter expert on enterprise introductions of new technologies as well as on immersive technologies, experiences and offerings.

APPENDIX B: PRESENTATION OF INFORMANTS 2(2)

Informant F

Informant F is based in Sweden and is working toward the future of smart and sustainable cities. Informant F is employed at a company working with smart city solutions. In this study, Informant F is considered a decision-maker in a company impacted by the introduction of immersive technologies.

Informant G

Informant G is based in Sweden, is a former academic, a research scientist at the Royal Institute of Technology and currently holds an R&D position at Ericsson. Informant G specialises in the human-machine interface, self-driving and connected vehicles and the optimisation of end-to-end mobile communication systems. Informant G contributes to this study as a subject matter expert with a particular interest in digital twins.

Informant H

Informant H is a postdoctoral research fellow. Informant H is leading the work at the Royal Institute of Technology on smart urban metabolism including data management and digitalisation to understand the flows of the city. Informant H is based in Sweden and has been part of multiple projects including both XR and digital twin technologies for developing smart and sustainable cities. Informant H is in this study considered a subject matter expert on smart cities with insights into the use of digital twins.

Informant I

Informant I is the CEO of a company developing AR technology for combining the digital and physical worlds. Informant I is based in Hong Kong and has previously worked in the telecom industry and within the IT software industry. Informant I contributes to this study as a subject matter expert on XR and immersive technologies.

APPENDIX C: OVERVIEW OF ANALYSIS 1(1)

Below, the illustration gives an overview of the analysis conducted according to the Gioia model with the 22 identified second-order themes to the left and how they relate to the four aggregated dimensions to the right.

Second-Order Themes	Aggregated Dimensions
New job roles	Individual
New skill sets	
Evolve with the change	
Technology is not really mature	
Focus on the end-user	
New ways of working	Company
Data security	
Regulatory factors	
Change journey	
Scaling	
Quantify the value	
Increase the value of offerings	
System functionality approach	Industry
Platform-thinking	
Unexpected origins of change	
Implementation drives adoption	
Portfolio for showcasing possibilities	
Industry cycles	
Fun and experimenting	New Innovation Processes
Culture	
New questions to ask	
New ways of inventing	

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